

From Liquidation Geometry to Price Movement

A Spatial-Density Heatmap and Cascade-Probability Model
for Cryptocurrency Perpetual Futures

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June 2026

Abstract

Leveraged perpetual-futures positions carry deterministic liquidation prices; when price reaches a dense cluster of them, forced market liquidations impose price impact that can reach the next cluster, producing a self-reinforcing cascade. We formalize this intuition. We construct a *spatial liquidation-density field*—a “liquidity heatmap”—that discretizes the log-return axis around the mid-price into $K = 200$ bins (10 basis points each, $\pm 10\%$) and maps the USD notional of each tracked position to the bin containing its liquidation price. From this field we extract eight scale- and level-invariant geometric features (nearest-cluster distance, mass concentration, cascade-chain length, directional asymmetry, absorption capacity, cluster velocity, mass-weighted distance, and spatial entropy). We then give a structural price-movement model in which a closed-form domino condition, $\delta_B - \delta_A < \lambda M_A/m_t$, determines whether liquidating one cluster triggers the next, with order-book depth acting as a damper (a liquidity spiral) and the directional asymmetry of the field predicting cascade sign. The geometry feeds an online, delayed-target logistic model of cascade probability. The contribution of this paper is the construction and the model, together with an explicit, falsifiable five-gate validation protocol and a candid treatment of the binding practical obstacle: on a decentralized venue the field is observed only through a *sample* of tracked wallets, not the full population of open interest. We formalize this sample-to-population bias, propose an open-interest rescaling that preserves field *shape*, and report the failure mode it produces in practice. We deliberately present no fabricated performance numbers; empirical validation under the stated protocol is the explicit next step.

Keywords: market microstructure, liquidation cascade, price impact, perpetual futures, spatial density estimation, fire sales, order-book liquidity, cryptocurrency

JEL Classification: G12, G14, G17, C53

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1 Introduction

1.1 The problem

Perpetual futures are the dominant instrument in cryptocurrency derivatives, and they are traded with high leverage. A leveraged position has a deterministic *liquidation price*: the level at which the maintenance-margin constraint binds and the exchange force-closes the position with a market order. These forced orders are price-takers and price-movers. When the mid-price approaches a region of the price axis that is dense with liquidation prices, a move into that region triggers a burst of same-signed market orders whose impact can carry the price into the *next* dense region—a liquidation cascade, the cryptocurrency analogue of a fire sale.

Two facts make this mechanism tractable as a microstructure object. First, the trigger levels are not latent: a position’s liquidation price is a deterministic function of its entry, leverage, and side. Second, the impact channel is the classical one: forced liquidations consume order-book depth, and their price impact is governed by market illiquidity. The open question is therefore not *whether* liquidation geometry matters, but how to *represent* it as a field over the price axis and how that field’s geometry maps to future price movement.

1.2 Approach and contributions

We represent the liquidation landscape as a one-dimensional density field $H(t, \cdot)$ over the log-return axis relative to the current mid-price, updated at 100 ms resolution (Section 3). From the field we extract eight geometric features that are invariant to price level and, being ratios or distances, approximately invariant to the unobserved scale of total open interest (Section 4). We then give a structural model of how the field moves price (Section 5): a single cluster’s impact follows Kyle (1985), a closed-form condition determines whether it triggers the next cluster, order-book depth damps the chain in the manner of Brunnermeier and Pedersen (2009), and the field’s directional asymmetry predicts the sign. The geometry drives an online logistic model of cascade probability with delayed targets (Section 6), gated by an explicit five-criterion validation protocol (Section 7).

Specifically, this paper contributes:

1. A construction of the liquidation-density field in *log-return* coordinates, making bins symmetric in percentage terms and invariant to price level, with an $O(N_t)$ update.
2. Eight scale-invariant geometric features of the field, each with a microstructural interpretation.
3. A structural price-movement model with a closed-form domino-trigger condition and an order-book absorption damper, linking field geometry to cascade magnitude and direction.
4. An online, delayed-target cascade-probability model and a falsifiable five-gate validation protocol.
5. A formal treatment of the sample-to-population observation problem on decentralized venues, an open-interest rescaling that preserves field shape, and an honest account of the coverage-driven failure mode observed in practice.

We emphasize at the outset that this is a *methods and model* paper. The geometry and the model are fully specified and implemented; a full out-of-sample evaluation under the protocol of Section 7 is stated as future work, and we report no measured performance figures we cannot substantiate.

2 Related Work

Price impact of forced trades. Kyle (1985) gives the canonical linear price-impact model in which order flow q moves price by λq ; λ is the inverse market depth. Forced liquidations are exactly such order flow. Our single-cluster impact and the domino condition are built directly on this primitive, with λ estimated from recent trades in the spirit of Amihud (2002) and Hasbrouck (2009).

Fire sales and endogenous risk. The cascade mechanism is a fire sale in the sense of Caccioli, Shrestha, Moore, and Farmer (2014), who analyze contagion in overlapping portfolios, and of Cont and Wagalath (2016), who measure endogenous risk created by feedback between forced selling and price. Our cascade-chain feature is the length of the triggered component of exactly such a contagion graph.

Liquidity spirals. Brunnermeier and Pedersen (2009) show how market and funding liquidity interact to amplify shocks in thin markets. Our absorption term—order-book depth divided by cluster mass—is the discrete, level-local form of their damper: a deep book absorbs a liquidation without propagation, a thin one lets price gap through.

Cryptocurrency liquidations. Perpetual-futures liquidation cascades are a recognized empirical phenomenon in crypto markets, but they are typically studied *ex post* from exchange liquidation feeds. Our contribution is a *forward-looking, spatial* representation built from position data, and an explicit treatment of the fact that on a decentralized venue this data is sampled, not complete.

3 The Liquidity Heatmap: Construction

3.1 Inputs and coordinates

Let m_t denote the mid-price at time t . The system tracks a set of leveraged positions; position i has notional V_i (USD), liquidation price ℓ_i , side, and entry price. We place each position on

the *log-return axis* relative to the mid,

$$\delta_i = \ln \frac{\ell_i}{m_t}, \quad (1)$$

so that δ_i is the (signed) percentage distance of the liquidation level from the current price. The axis is discretized into K uniform bins on $[-R, R]$:

$$\delta_k = -R + (k + \frac{1}{2})\Delta, \quad \Delta = \frac{2R}{K}, \quad \mathcal{B}_k = (-R + k\Delta, -R + (k+1)\Delta), \quad k = 0, \dots, K-1, \quad (2)$$

with defaults $K = 200$, $R = 0.10$ (so $\Delta = 0.001$, i.e. 10 basis points per bin, spanning $\pm 10\%$). Working in log-returns makes the bins symmetric in percentage terms and independent of the absolute price level.

3.2 The density field

Definition 3.1 (Liquidation-density field). The liquidity heatmap is the field

$$H(t, k) = \sum_{i=1}^{N_t} V_i \mathbf{1}[\delta_i \in \mathcal{B}_k] \mathbf{1}[V_i \geq \tau_V], \quad k = \left\lfloor \frac{\delta_i + R}{\Delta} \right\rfloor, \quad (3)$$

i.e. $H(t, k)$ is the total USD notional whose liquidation price lies in bin k . Positions with notional below τ_V (default \$1,000) or with $\delta_i \notin [-R, R]$ are discarded.

The field is rebuilt in full each snapshot; each position maps to one bin via a logarithm, an addition, and a floor, so the update is $O(N_t)$. There is no temporal smoothing of H itself; temporal information enters only through the velocity feature below.

3.3 Derived channels

Three further channels give the field local structure (here $k_0 = K/2$ is the at-the-money bin):

$$\text{cumulative mass: } C(t, k) = \sum_{j \text{ between } k \text{ and } k_0} H(t, j), \quad (4)$$

$$\text{cluster gradient: } G(t, k) = H(t, k) - H(t, k-1), \quad (5)$$

$$\text{temporal change: } T(t, k) = \alpha_1 \frac{H(t, k) - H(t - \Delta t_1, k)}{\Delta t_1} + \alpha_2 \frac{H(t, k) - H(t - \Delta t_2, k)}{\Delta t_2}, \quad (6)$$

with $\Delta t_1 = 600$ ticks (1 min), $\Delta t_2 = 3000$ ticks (5 min), and weights $\alpha_1 = 0.7$, $\alpha_2 = 0.3$. The cumulative channel measures the fuel between the mid and a level (cascade reach), the gradient locates cluster boundaries, and the temporal channel is the cluster-migration velocity field.

4 Spatial Features

Eight scalar features summarize the field's geometry each snapshot. Write the total mass $\Sigma_H = \sum_k H(t, k)$, $\bar{H} = \Sigma_H/K$, and let $p_k = H(t, k)/\Sigma_H$ be the normalized mass. Let $\tau_{cl} = \$10^6$ and $\tau_{ch} = \$10^5$ be the cluster and chain thresholds.

These features are deliberately constructed to be *ratios or signed distances* (Table 1): CMR, ACP, S , $\bar{d}$, and $d_{\min}$ are unchanged if the entire field is rescaled by a common factor. This is the property that makes them usable when only a fraction of total open interest is observed (Section 8).

Feature	Definition	Interpretation
hm_nearest_cluster_distance	$d_{\min} = \min\{ \delta_k : H(t, k) > \tau_{cl}\}$	distance to the nearest large cluster (smaller = cascade closer)
hm_cluster_mass_ratio	$CMR = H(t, k^*)/\bar{H}, k^* = \arg \max_k H$	peak concentration relative to mean
hm_cascade_chain_length	longest run of consecutive bins with $H > \tau_{ch}$ within $ \delta \leq 0.05$	domino potential (triggered-chain length)
hm_asymmetric_cascade	$Aspt = \frac{\sum_{0 < \delta \leq .05} H - \sum_{-.05 \leq \delta < 0} H}{\sum_{ \delta \leq .05} H + \epsilon}$	directional bias: > 0 short-squeeze, < 0 long-cascade
hm_absorption_capacity	$A = \min(\frac{D^{bid}(.02)}{\text{down mass} + \epsilon}, \frac{D^{ask}(.02)}{\text{up mass} + \epsilon})$	book depth per unit liquidation mass (< 1 = insufficient damping)
hm_cluster_velocity	$v = (d_{\min}(t) - d_{\min}(t - \tau))/\tau, \tau = 600$	rate of approach of the nearest cluster (< 0 approaching)
hm_mass_weighted_distance	$\bar{\text{distance}} = \frac{\sum_k \delta_k H(t, k)}{\sum H + \epsilon}$	centre of liquidation gravity
hm_heatmap_entropy	$S = -\sum_k p_k \ln p_k$	spatial concentration (S low = tight clusters)

Table 1: The eight liquidity-heatmap features. ϵ is a small constant (1 USD). Absorption uses order-book depth within 2% on each side; when a side carries no mass its ratio is treated as $+\infty$, and A is clamped to $[0, 100]$.

5 The Cascade Mechanism: A Price-Movement Model

5.1 Single-cluster impact

Let cluster A at δ_A carry mass $M_A = H(t, k_A)$. Treating its forced liquidation as an order of notional M_A and applying Kyle (1985) with depth parameter λ (estimated from recent trades), its price impact in log-return units is

$$\Delta p_A = \lambda \frac{M_A}{m_t}. \quad (7)$$

5.2 The domino condition

A second cluster B at $\delta_B > \delta_A$ (on the same side of the mid) is reached if and only if cluster A 's impact closes the gap:

$$\boxed{\delta_B - \delta_A < \lambda \frac{M_A}{m_t}} \quad (8)$$

Equation (8) is the structural heart of the model. It depends on the *spacing* of clusters (captured by the gradient channel and the chain-length feature) and their *mass* (the concentration feature), mediated by illiquidity λ . When it holds across a run of clusters $\delta_1 < \dots < \delta_n$, the total move is the sum of triggered masses,

$$\Delta p_{\text{total}} \approx \frac{\lambda}{m_t} \sum_{j: \text{triggered}} M_j, \quad (9)$$

and the number of triggered nodes is exactly the cascade-chain feature, read off a Caccioli et al. (2014) contagion graph.

5.3 Absorption damping

Order-book depth opposes the cascade. With bid depth $D^{\text{bid}}(\delta_A)$ available to absorb a downward liquidation at A , define the damping

$$\eta = \min\left(1, \frac{D^{\text{bid}}(\delta_A)}{M_A}\right), \quad \Delta p_A^{\text{eff}} = (1 - \eta) \Delta p_A. \quad (10)$$

When $\eta \rightarrow 1$ (a deep “wall,” i.e. absorption $A \gg 1$) the impact is absorbed and the chain breaks; when $\eta \rightarrow 0$ (a thin book) it is amplified—the Brunnermeier–Pedersen (2009) spiral. This is the mechanism behind support/resistance walls and absorption.

Proposition 5.1 (Magnet and rejection). Under (8)–(10), a large cluster within reach acts as a price *magnet*: conditional on the mid entering its neighborhood, the probability of a move toward it increases with M_A and decreases with the local absorption A . Conversely, a cluster fronted by a deep book ($\eta \approx 1$) produces *rejection* rather than breakout.

5.4 Direction

The sign of the field’s near-money asymmetry predicts the cascade direction. With ACP as in Table 1, mass concentrated above the mid is short-liquidation fuel (an upward squeeze) and mass below is long-liquidation fuel (a downward cascade), so the predicted direction is $\text{sign}(\text{ACP})$.

6 A Probabilistic Cascade Model

6.1 Event label

We define a cascade over horizon T as a large *liquidation-driven* move—a price move conjoined with abnormal liquidation volume, which distinguishes it from an ordinary news move:

$$y_t = \mathbf{1}\left[|\Delta p_{t \rightarrow t+T}| > X\% \wedge \text{LiqVol}_{t \rightarrow t+T} > Y\right], \quad \Delta p = \ln \frac{m_{t+T}}{m_t}, \quad (11)$$

with defaults $X = 3\%$, $Y = \$500\text{K}$, $T = 3000$ ticks (5 min), giving a base rate near 5%. Because per-tick liquidation volume is not directly observable on the venue, the implementation currently uses the price-threshold component as a proxy—a conservative choice we flag explicitly.

6.2 Features and model

The model consumes an 11-dimensional vector: the eight features of Table 1 plus three interactions chosen from the mechanism,

$$\mathbf{x}(t) = \left[d_{\min}, \text{CMR}, L, \text{ACP}, A^{-1}, v, \bar{d}, S, \underbrace{L \cdot A^{-1}}_{\text{fuel, no damper}}, \underbrace{v \cdot \mathcal{I}}_{\text{approach} \times \text{illiquidity}}, \underbrace{\text{ACP} \cdot \text{sign}(f)}_{\text{aligned with funding}} \right]^\top, \quad (12)$$

where L is the chain length, $\mathcal{I}$ a composite illiquidity, and f the funding rate. Cascade probability is an online logistic regression,

$$P(\text{cascade} \mid \mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (13)$$

trained by stochastic gradient descent on the cross-entropy loss with ℓ_2 regularization ($\lambda_{\text{reg}} = 10^{-3}$). With per-step gradient $g_t = \hat{p}_t - y_t$,

$$\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} - \eta_t (g_t \mathbf{x}_t + \lambda_{\text{reg}} \boldsymbol{\beta}), \quad \beta_0 \leftarrow \beta_0 - \eta_t g_t, \quad \eta_t = \eta_0 \gamma^t, \quad (14)$$

with $\eta_0 = 0.01$, $\gamma = 0.9999$. Because the label (11) is only known T ticks later, updates are applied from a *delayed-target* buffer once each observation matures; inputs are standardized online by Welford’s algorithm, and the logit is clipped to $[-20, 20]$. A logistic model is deliberate: cascades are rare ($\approx 5\%$), leverage regimes are non-stationary, and the coefficients β are interpretable as the weight the data place on each geometric feature.

7 Validation Protocol

The model is only credible if it clears pre-registered gates. We specify five; they are *thresholds to be met*, not results claimed here.

Gate	Criterion
G1 — Discrimination	out-of-sample AUC > 0.65 and top-decile lift $> 2\times$
G2 — Net information	$IC_{\text{net}} = IC_{\text{gross}} - \frac{2\bar{s}}{\sigma_r\sqrt{T}} > 0.02$
G3 — Stability	walk-forward median AUC > 0.60 , $< 20\%$ of windows below 0.55
G4 — Universality	BTC, ETH, SOL each pass G1, and $\cos(\beta_i, \beta_j) > 0.6$ pairwise
G5 — Orthogonality	$R^2 < 0.30$ regressing $\hat{p}$ on a spread/depth composite

Here $\bar{s}$ is the cascade-event spread (a cost proxy) and σ_r the return volatility. G2 ensures the signal survives transaction costs; G5 ensures it is not a relabeling of ordinary liquidity. The prior empirical anchor that motivates the whole construction is the hypothesis-test result that coarse liquidation-cluster proximity predicts large moves with lift $> 2\times$; the heatmap’s claim is that its eight *spatial* features carry information *incremental* to that coarse proximity—a claim Gate G5 and an explicit ablation against `hm_nearest_cluster_dist` alone are designed to test.

8 Data Availability and the Sample-to-Population Problem

The model’s binding obstacle is observational, not theoretical. On a decentralized venue, position data is obtained by querying the clearinghouse state of *known wallet addresses*; no endpoint returns the full population of open positions. Wallets are discovered from the public trade stream and polled, yielding coverage on the order of 50–500 wallets per symbol—an *unknown* fraction of open interest. The field of Definition 3.1 is therefore a sample, $H^{\text{obs}}(t, k) = \rho_t H(t, k) + \text{noise}$, with sampling rate ρ_t unobserved and time-varying.

What survives sampling. The scale-invariant features— $d_{\min}$, CMR, ACP, $\bar{d}$, S —depend on the field’s *shape*, which is preserved in expectation under uniform sampling; and the largest, cascade-causing positions are disproportionately likely to be discovered (they dominate trade flow). What is biased are the *absolute-mass* quantities: absorption capacity A is overestimated (observed mass underestimates the true denominator), and chain length may be undercounted when untracked positions fill gaps in (8).

A shape-preserving rescaling. Because total open interest *is* observable, we can restore scale without distorting shape,

$$H^{\text{scaled}}(t, k) = H^{\text{obs}}(t, k) \cdot \frac{OI_{\text{total}}(t)}{\sum_k H^{\text{obs}}(t, k)}, \quad (15)$$

which leaves every ratio feature unchanged while correcting the mass-dependent ones. Validating (15) against realized cascade magnitudes is itself a clean empirical sub-study.

Observed failure mode. We report candidly that, in a preliminary multi-symbol backtest, the cascade-probability algorithm produced *zero* trades and the nearest-cluster-distance feature frequently returned its no-data sentinel—direct evidence that, under current wallet coverage, the observed field is too sparse for the absolute-mass features to fire. This is the single most important practical result of the work to date: the geometry and the model are sound, but their utility is gated on coverage, and the first priority is therefore data—aggressive wallet discovery and the rescaling (15)—not further modeling.

9 Discussion

The model makes several falsifiable claims that a venue with adequate coverage could test directly. The domino condition (8) predicts that the probability of a multi-cluster cascade jumps discontinuously as inter-cluster spacing falls below $\lambda M_A/m_t$; the absorption relation (10) predicts that deep near-money book depth converts approach into rejection rather than breakout (Proposition 5.1); and the directional claim predicts that $\text{sign}(\text{ACP})$ leads cascade sign. Each is a clean hypothesis with a pre-specified statistic. The most important open question is incremental information: do the eight spatial features add predictive content beyond simple cluster proximity? The construction is designed so that this question is answerable by ablation, and Gate G5 guards against the field being a disguised liquidity proxy.

The honest status is that this is a fully specified instrument awaiting the data to validate it. That ordering—specify the mechanism, expose the observational constraint, fix the data, then measure—is, we argue, the correct one for a phenomenon whose theoretical structure is clear but whose measurement on a decentralized venue is genuinely hard.

10 Conclusion

We have constructed a spatial liquidation-density field for cryptocurrency perpetual futures, extracted eight scale-invariant geometric features from it, and given a structural price-movement model in which a closed-form domino condition, an order-book damper, and a directional asymmetry map the field’s geometry to cascade magnitude, occurrence, and sign. The geometry drives an online, delayed-target cascade-probability model, gated by an explicit five-criterion protocol. We have been deliberate in not reporting performance numbers we cannot substantiate, and in foregrounding the binding constraint: on a decentralized venue the field is sampled, not observed in full, and the immediate priority is the coverage and rescaling needed to make the absolute-mass features fire. With that data in hand, every claim above becomes a pre-registered, testable hypothesis.

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